

Information apocalypse or overblown fears—what AI mis- and disinformation is all about? Shifting away from technology toward human reactions

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Abstract

The rise of generative artificial intelligence (AI) has ignited a debate about its effects on the mis- and disinformation landscape. The doomsday scenarios of epistemic and information apocalypse presented for many years are recently being questioned, and the previous fears are called “overblown.” These phenomena are analyzed mostly through the factors of quantity and quality of AI-powered content and the potential for personalization possessed by AI. We argue that using quantitative arguments carries a high risk of underestimating the threat, especially in the context of the so-called detection challenge. We point out that this discourse is affected by the narrow conceptualization of how we understand quantity, quality, and personalization with regard to AI. In our opinion, apocalyptic visions are speculative in nature, difficult to quantify, and carry signs of a self-fulfilling prophecy, but disregarding risks hinders appropriate countermeasures against AI-powered dis- and misinformation, which adversely affects policy-making activities. We propose a paradigm shift to focus more on social reactions to technology rather than technological attributes. By expanding the understanding of the analyzed phenomena, we indicate that the potential of AI is both overestimated and underestimated and above all—still misunderstood.

KEYWORDS

AI, AI-powered mis-information, artificial intelligence, chatbots, deepfakes, deepfakes and legal evidence, detection software, fake news, human psychological reactions to AI, information apocalypse, information manipulation, mass-scale disinformation, personalized political campaigning, risk, social media, strategic interpretation of

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AI, technology policy, trust, U.S. 2024 election, uncertainty

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The rise of generative artificial intelligence (AI) has ignited a debate about its effects on the mis- and disinformation landscape. Some scholars foresee doomsday scenarios of epistemic and information apocalypse (Fallis, 2021; Schick, 2020), AI being used as a "weapon of mass disruption" (Bremmer & Kupchan, 2023), or a complete blurring of boundaries between true and false (Metz, 2023). Recent voices, on the other hand, claim AI is merely an "evolution, not a revolution" (Muñoz, 2024), pushed not to focus on a single technology (Bateman & Jackson, 2024) when it comes to disinformation, or call fears about AI-misinformation "overblown," pointing to a lack of empirical evidence of AI-supercharged mis- and disinformation as well as an already well-functioning "marketplace of misinformation" that fulfills both providers' and consumers' needs (Simon et al., 2023).

While an "information apocalypse" has certainly not (yet?) materialized, we argue that the recent past has seen significant and increasing cases of AI-powered mis- and disinformation with real-life consequences. Even if apocalyptic visions of AI-powered mis- and disinformation might be exaggerated, disregarding the risks is not justified, especially since psychological reactions to AI-powered mis- and disinformation already play an important role in shaping attitudes and political actions. Therefore, we suggest a shift in previous paradigms and focus more on human reactions to AI-powered mis- and disinformation and to the mere, although sometimes only imaginary, potential of AI technology. It turns out that technology does not even have to be used to have a specific impact on the information environment and society, as its sheer existence leads to the erosion of trust and increased uncertainty among recipients (Twomey et al., 2023; Vaccari & Chadwick, 2020).

METHODOLOGY AND THE CONCEPT OF THE STUDY

Recognizing these trends, we draw specifically upon a valuable and thought-provoking study by Simon and others (2023) on AI misinformation titled "Misinformation reloaded? Fears about the impact of generative AI on misinformation are overblown." In the present piece we focus on highlighting a number of elements that may be insufficiently recognized by the authors of this study and add another dimension that should be taken into account to assess the real potential of AI-powered mis- and disinformation. Therefore, for methodological reasons, our analysis mirrors the structure of the study by Simon and others (2023). Its authors focused



on three main areas regularly recognized by scholars, and thus assessed the potential of AI to increase the quantity, quality, and personalization of misinformation, and criticized narratives which in their opinion inadequately overestimate the threat.

A narrative built in direct opposition to apocalyptic visions may result in underestimating threats and challenges. Our analysis should not be understood as a fundamental criticism of the shift away from alarmist narratives about the effects of AI. We agree with the basic thesis that some fears are indeed “overblown” (Simon et al., 2023), and they stem from a journalistic discourse (Łabuz & Nehring, 2024), but we also suggest a more nuanced approach that would be beneficial for policy makers and discourse at large. Simon and others (2023) stipulate that their revised approach should not be understood as simply dismissing concerns around the technology, and this remark should be assessed positively. Their arguments, therefore, might be a starting point for further discussions. Scholars at large rightly emphasize the need to conduct better quality research and not go to extremes. Therefore, the identified factors such as quantity, quality, and potential for personalization are, on the one hand, too few to assess the effects of AI-powered mis- and disinformation, and on the other hand, their narrow conceptualization makes it difficult to estimate the actual scope of the phenomenon, which, in our opinion, at least partially distorts the discourse.

We argue first that, due to the “detection challenge,” it remains impossible to achieve an actual quantitative assessment of AI-powered mis- and disinformation. Second, we present ample empirical evidence for the existence of AI-generated fakes and manipulative content and their improving quality that results in severe consequences for individuals, groups, and ultimately societies. Third, we delve into the ongoing debate about AI's potential for supercharging personalized and customized political messages and manipulations, pointing toward new experimental studies suggesting that AI-generated personalized content is more effective than non-personalized. We also argue with a very narrow conceptualization of how we should understand personalization. Thus, we propose a shift from focusing primarily on AI's technological capabilities, such as quantity, quality, or potential for personalization (Simon et al., 2023), to its psychological and social effects on human audiences. We argue that AI mis- and disinformation already achieves significant effects due to the way human audiences perceive it and react to it. Moreover, it also has a direct impact on social and political processes, although in a different way than it was expected some years ago.

At the same time, a limited understanding of quantity, quality, or personalization could reduce the scope of analysis, distorting the picture of AI-powered information disorders, which have been considered the greatest global challenge in the coming years by the World Economic Forum (2024). The qualification itself may contribute to a spiral of fear.

It should be emphasized that Simon and others (2023) consistently use the term “misinformation,” which within the methodological framework of HKS Misinformation Review refers to “false or inaccurate information—regardless of whether it is shared with the intention to deceive” (HKS Misinformation Review, 2024). Therefore, it is supposed to cover both misinformation and disinformation. Our approach to defining mis- and disinformation is more conservative, and we separate both notions in terms of their meaning—in the case of disinformation, we take into account the aspect of intentional dissemination of false or misleading content. However, paying attention to the original formulations used by Simon and others (2023), we generally use the collective term “mis- and disinformation,” which takes into account a broader conceptual scope of different information disorders and should be seen as a direct reference to the commented study by Simon and others (2023). In the case of the personalization aspect, we exclusively use the term disinformation to clearly indicate what this phenomenon refers to.



CRITICAL ASSESSMENT OF THE ONGOING DEBATE

The debate on AI mis- and disinformation has been shaped by two conflicting paradigms: some scholars have promoted pessimistic narratives about AI's potential to supercharge information disorders to such an extent that it poses an existential threat to the global information space and the “epistemic value” of audio and visual content (Fallis, 2021; Gold & Fischer, 2023; Goldstein et al., 2023; Ienca, 2023; Metz, 2023; Rini, 2020; Schick, 2020). These concerns replicated patterns that have accompanied emerging technologies over the centuries (Murphy et al., 2023). They were partly justified, especially with the advent of technology that allows for the creation of increasingly convincing deepfakes; i.e., AI-generated or AI-manipulated audio, image, or video content that resembles existing persons, objects, places, or other entities or events and falsely appear to a person to be authentic or truthful (AI Act, 2024). Since its appearance in 2017, deepfakes have been consistently considered a threat to democracy and society (Farid & Schindler, 2020; Vaccari & Chadwick, 2020). Several studies have reflected fears that deepfakes lead to a disruption of the epistemic value of audio and visual content, whereas the level of uncertainty as to the authenticity corresponds to a growing lack of trust in news and the media as such (Home Security Heroes, 2023; Luminare, 2023; Twomey et al., 2023).

One weakness of this apocalyptic narrative, deeply rooted in journalistic discourse, was the lack of quantification of the described phenomena (Łabuz & Nehring, 2024). It is difficult to give a detailed estimate at what point an epistemic apocalypse would occur, even if the observations of social and psychological reactions to deepfakes reveal some negative trends. Furthermore, this pessimistic narrative rests on future predictions about possible scenarios of AI-powered mis- and disinformation, which are built upon the technological possibilities inherent to generative AI. In addition, the display of terrifying visions of AI-supercharged information disorders, intended to be an element of building awareness, could fuel a spiral of fear and reduce trust in the credibility of the media, acting as a self-fulfilling prophecy.

The demonization of AI creations and consistent warnings about the devastating effects of information disorders have caused other scholars (Immerwahr, 2023; Murphy et al., 2023; Simon et al., 2023) to start taking a critical look at selected factors or to draw attention to the speculative nature of these considerations. However, areas identified by Simon and others (2023) relating to the quantity, quality, and personalization potential of AI mis- and disinformation refer only to some of the negative attributes of the phenomenon and disregard recent research results as well as the pure practice of using malicious tools, especially in the case of criminal activities. It appears that these arguments, while correctly questioning the manifestation of doomsday scenarios, fail to take into account several important aspects—which we highlight below—thus reaching an incomplete conclusion.

SIGNIFICANCE OF THE DETECTION CHALLENGE

Simon and others (2023) pointed out that there is no empirical evidence to prove that AI already has a significant effect on the quantity of misinformation since existing mis- and disinformation ecosystems do not need AI supercharging to meet demand and supply. In some respects, this argument is supported by numerous scientific studies, proving that demand for disinformation develops in specific environments and is correlated with specific personality traits (Motta et al., 2023). The factors contributing to its dissemination include high polarization and low trust in the media (Jenke, 2023; Osmundsen et al., 2021). Similar relationships were studied by Oyserman and Dawson (2020) and Diaz Ruiz and Nilsson (2023) pointed to identity-based factors that increase susceptibility to disinformation and the creation of the so-called echo chambers. The consumption of disinformation content might, therefore, often be limited to specific environments populated by individuals with specific personality traits.



The counternarrative to doomsday scenarios is mostly technology oriented with quantitative and qualitative approaches focusing on the technological capabilities of AI. However, there are neither technological nor other capabilities that would allow for even a rough estimation of the quantity of existing AI-powered mis- and disinformation today (Sadasivan et al., 2023). The methodological frameworks are based on unreliable mechanisms, including content moderation or detection software characterized by low effectiveness at this stage of technological development. Proposals for changes in this area must take into account the difficulties in detecting AI-powered content (Karinshak & Jin, 2023). At the policy-making level it might result in implementing technical solutions based on watermarking, content provenance, or authentication. Some of these solutions are already being tested and even implemented as part of the regulatory framework for AI, which is the case, for example, with the proposals included in the EU AI Act (2024). This gives hope for slightly better estimates, although they will still be at a high risk of underestimation.

The so-called “detection challenge” is of relevance for any evaluation and risk assessment of AI mis- and disinformation. As studies have shown, text produced by large language model (LLM) bots can still not be reliably detected with detection software, which either shows too little accuracy or can easily be tricked by automatic rephrasing or translation (Lucas et al., 2023). Furthermore, while public discourse on AI threats focuses on popular AI applications, such as ChatGPT or Gemini, it is uncertain whether professional actors of mis- and disinformation would rely (only) on them since they have at their disposal a high number of freely available open-source LLM bots that are easily customizable and virtually undetectable (Tang et al., 2023). Problems with the automatic detection and labeling of AI-generated content by digital platforms seem to confirm these trends.

The situation with deepfakes is only slightly better. Even though there is an increasing number of deepfake detection tools available, they also suffer from insufficient accuracy, producing “false positives” (i.e., falsely labeling content as AI produced) and “false negatives” (i.e., falsely labeling content as not AI produced; Dolhansky et al., 2020; Le et al., 2023). Thus, despite the detection software for AI-generated or AI-manipulated audio and visual content that works better than text-detection software, some deepfakes are still being identified by comparing data or debunking flaws in images and videos produced by AI. These gradually disappear as the quality of materials increases, which poses a significant threat to content moderation, regardless of its form. Again, security measures at the model creation stage (including watermarking or marking metadata) and obligations toward deployers may eliminate some of the problems, but they will not constitute an insurmountable barrier for professional state and nonstate actors.

In light of this “detection challenge,” the question of how we can determine the actual quantity of AI-produced mis- and disinformation remains. It seems that without knowing or even being able to estimate the real quantity of AI mis- and disinformation, we can hardly support the claim that there is no empirical evidence for AI that supercharges mis- and disinformation.

IS THERE AI MIS- AND DISINFORMATION?

While the detection challenge makes it impossible to accurately assess the quantity of AI mis- and disinformation, there is nevertheless ample empirical proof for its existence. Some researchers have come up with creative (yet nonrepresentative) approaches to prove that fake content generated by LLMs is already being produced and spread. For example, searching social media for some of the give-away phrases produced by LLMs when they cannot accurately fulfill a task (e.g., “as an AI model, I cannot...”) has revealed bot-networks trying to involve users in financial scams on “X” (Yang & Menczer, 2023); thousands of fake reviews and comments on Amazon, Google, Yelp, etc. (Dekens, 2023); numerous online portals rewriting old



news articles for clickbait (Brewster et al., 2023); or as a part of alleged Russian webs of mushroom sites (Linville & Warren, 2023). These examples of using AI text generators to produce mis- and disinformation are far from being representative and they cannot be used for an assessment of the real quantity of AI-powered mis- and disinformation too, yet they clearly prove its existence and most probably point to the tip of the iceberg of AI-generated fake-content underneath. Interestingly, in most of these cases, it is impossible to evaluate the reasons why disinformation actors chose to use AI to create their content or whether they used AI simply because they could.

Recent months have also seen a rapid increase in the production and spread of AI images as well as deepfakes. Some examples include AI images to bolster fake social media accounts (Yang et al., 2024), AI images that supported Chinese campaigns spreading conspiracy theories about wildfires in Hawaii (Sanger & Myers, 2023), the propaganda war following Hamas' attack on Israel (Maiberg, 2023a), or popular images posted on social media being edited with AI for clickbait (Koebler, 2023). These materials were met with a great interest from public opinion, which resulted in amplification by sharing and hundreds of thousands of views. In May 2023, a deepfake depiction of an explosion near the Pentagon was spread on social media and led to fluctuations in the New York Stock Exchange. Bloomberg called it "possibly the first instance of an AI-generated image moving the market" (Hurst, 2023). All these cases were met with great interest from the recipients, although in some cases, the reach was limited to specific groups, which fits into the patterns discovered in the case of classic forms of information disorders, without the participation of AI (Diaz Ruiz & Nilsson, 2023).

Our goal here is not to verify the effectiveness of the analyzed tools, but to refer to measurable and nonmeasurable values. At the same time, a very recent study from Goldstein and others (2024, p. 5) proved that AI-generated propaganda can be nearly as persuasive as human-generated professional content and "human-machine teaming strategies (editing prompts and curating outputs) produced articles that were as, or more, persuasive than the original propaganda." Spitale and others (2023) indicated that disinformation generated by GPT-3 is more compelling. These are just two selected examples where the increasing quality of synthesis translates into greater capabilities for disinformation. Paying attention to the human-machine teaming strategy shows that appropriate supervision can increase persuasiveness and at the same time reduce costs (Goldstein et al., 2024).

Experience with the spread of AI-powered mis- and disinformation and the amplification of this type of content by social media algorithms shows that the demand is large enough to lead to several dozen million views of a deepfake pornographic video featuring the singer Taylor Swift (Moench, 2024), or tens of millions of interactions with audio deepfakes generated by AI for "a network of TikTok accounts posing as legitimate news outlets" (Murphy, 2024). In the first case, the demand was so huge that it forced "X" to completely block the ability to search for Swift's name.

The example of the campaign before the presidential election in Argentina in October 2023, where deepfakes and other AI content were produced on a mass scale and the information space was literally flooded with them, shows that public opinion is ready to engage with the AI content of a political nature and experiment on its own with new creations (Nicas, 2023). We do not judge whether these interactions are mostly conscious actions or not. However, studies have proven that even if users are aware of their exposure to fake content, the consequences can be severe, strengthening cognitive bias and affecting associations with the depicted objects in the future (Brown, 2020)—which in itself should be a warning sign against disregarding superficially harmless forms of experimenting with AI-generated content by public opinion.



THE TRAPS OF MISUNDERSTANDING AI PERSONALIZATION

AI's ability to customize and personalize content is another popular argument for its game-changing capabilities when it comes to disinformation (Farid, 2022). Opposing views, however, do not see sufficient empirical evidence for an actual effect of personalized political messaging with or without AI (Newport, 2023; Simon et al., 2023; Zarouali et al., 2022), thus questioning the need for personalized AI messaging. Efficiency may be a key factor here. Even though there is no empirical evidence that AI produced, highly convincing, personalized political advertising that made a decisive difference in any campaign (Łabuz & Nehring, 2024), there is ample evidence to prove that AI is nevertheless being used to personalize political campaigning. Moreover, very recent studies in the field of cognitive sciences (Simchon et al., 2024) have produced empirical data that proves that personalization of AI messaging indeed works, though its effectiveness is only slightly higher. It has also been proven that the use of appropriate mimicry or facial resemblance with the recipient positively influences the attitude toward presented persons, or enhances the persuasion (Bailenson et al., 2008; Olszanowski et al., 2022).

It likewise appears that the discussion about the personalization potential of AI is strongly anchored in a rather stereotypical perception of what personalization might look like. Here, it seems that this misunderstanding may be rooted in the difference between what observers expected it to be and how AI is used in political reality. Instead of highly personalized aggressive voting recommendations, reality most often sees other, quite different use cases: already in 2020, the campaign of Indian politician Manoj Tiwari used deepfakes enabling him to address his speech in a dialect of Hindi, which he does not actually speak. It reached an audience of 15 million citizens (Nilesh, 2020). Yoon Suk-yeol won the 2022 South Korean presidential election, and along the way, he successfully used deepfakes to generate interest in his campaign among the youngest voter age group (Martin & Yoon, 2022). In India again, campaigners have experimented with personalized deepfake-WhatsApp messages with the use of recipients' names and real-time voice translation (Christopher, 2023). Recent US election campaigns have seen several cases of generative AI being used for automated phone calls, e.g., New York mayor Eric Adams using robocalls with AI translation to minority voters he otherwise could not have reached (Simon, 2023), or Shamaine Daniels' use of an AI bot for automated, fully AI operated one-to-one campaign phone calls with voters (Tong & Coster, 2023). Joe Biden's voice was used for a massive robocall campaign discouraging voters from particular areas from participating in New Hampshire's presidential primary (Murphy, 2024). These are just selected examples where AI was used for sophisticated manipulations of a political nature.

Meanwhile, content personalization may also include, for example, AI-generated pornography aimed at using personalized visualizations, including child sexual abuse materials (CSAM). The consequences are severe, leading to the additional victimization of vulnerable groups, multiplying CSAM circulating online, thus impeding the work of law enforcement agencies, and increasing threat perception (IWF, 2023). Moreover, regularly reported cases of "deep porn" have affected the collective sense of security and added another building block to gender inequality, targeting mainly women (Okolie, 2023). Also in this respect, it should be emphasized that statistics on deepfakes circulating in the information space indicate significant increases in the generated content—in comparison to 2019, five times more deepfake videos were registered in 2023 (Home Security Heroes, 2024). The vast majority of them concern deep porn, the social harmfulness of which is still underestimated and does not receive an effective response from policy makers and digital platforms.

These examples may not reflect what both sides of the debate expected personalized AI content to look like. Whoever expected a massive amount of personalized voting recommendations automatically produced by LLMs with the press of a button or deepfake avatars speaking directly and personally to each voter with a convincing, customized, and personalized message,



might have overestimated AI's capabilities as well as audience and campaigners' reactions to it. Such forms of hyper-personalized manipulative deepfakes are, however, currently gaining track in organized crime and scamming attacks. In Germany, for example, law enforcement investigated a wave of audio deepfakes targeting individuals with frightening messages about relatives being arrested, abducted, or hospitalized in order to extort money (Rödger, 2023). In other cases of financial fraud, audio deepfakes were used in combination with social engineering and cyber attacks to trick company accountants into transferring large amounts of money (Yu, 2024). In 2023 Onfido (2024) registered a 31× increase in the number of deepfake fraud attempts in comparison to 2022. Further reports on the effectiveness of new fraud schemes show that personalization indeed works, and AI has become an instrument of cybercriminals' activities.

These are specific threats that affect the social sense of security and stability, attacking not only individual recipients of AI-generated personalized content but also collective values and perception. The inability to notice them may be due to too narrow a conceptualization of how we understand personalization.

FINITE ATTENTION, INFINITE SUPPLY

Simon and others (2023) refer to the interesting argument that “the number of things that go viral on the internet and get seen is finite because our attention is finite.” Indeed, our capacity is finite, but the development of new forms of communication, primarily social media, has disrupted traditional interactions with sources. It also disrupted the parity between verified and proven news and those that distort reality. Our attention may be finite, but it is based on what is available. If AI could bombard us with false or manipulated information, then human capacity could not play such an important role.

Moreover, our interactions with mis- and disinformation do not have to be unconscious in this respect. A significant part of the audience must have realized that the deep porn depiction of Taylor Swift was fake. Despite this, they were willing to interact with the content and thereby enhanced algorithmic amplification. The example of deep porn with Swift should be a valuable lesson in this respect and, we hope, a turning point in the assessment of a phenomenon that affects thousands of victims, mainly women, around the world.

The multiplication of fake content on a mass scale has shown that the pace of the spread is too fast for an effective response by digital platforms, and the interest of users is great enough to drive further supply. In addition, arguments about finite attention seem to disregard the fact that not all mis- and disinformation content is meant to be persuasive. Instead, coordinated campaigns are quite often meant to demobilize engagement and divert attention, to repel audiences from debates, processes, and topics by flooding online information spheres with low-quality content. All this is supposed to have a specific impact on the cognitive abilities of the community. Microsoft warned, for example, against the *spamouflage* strategy used by actors associated with China, drawing attention to the significant increase in the use of AI to operate influence campaigns and flooding adversaries' information space with AI-generated disinformation content. It has also been expressed in the multiplication of AI-powered social media accounts that enhance the spread of false or misleading messages (Microsoft Threat Intelligence, 2024).

Concerning the impact of AI fakes on the political process, the effects of AI-powered mis- and disinformation are not clearly visible, which makes it hard to see doomsday scenarios coming true. This does not mean they do not exist. For example, in Turkey a candidate had to withdraw from running in the presidential elections due to a discrediting deep porn video spread online a couple of weeks before voting (Michaelson, 2023). In Slovakia and Bangladesh, attacks on the integrity of elections were made just before the vote, targeting so-called



“decisional checkpoints” when there is too little time to debunk false content (Chesney & Citron, 2019), though the outcome was mediocre and it did not significantly change the results of the election. There are many examples of politically motivated deepfakes, and many of them attract great public interest, receiving millions of views, such as the deepfake audio with the UK opposition leader Keir Starmer, swearing and abusing his coworkers (Bristow, 2023). They are also shared by public figures who are unaware of the fake nature of the content (Mishra & Karumbaya, 2024), thus increasing their reach and enhancing algorithmic amplification, which might be seen as another dimension of AI boosting mis- and disinformation and personalization.

The mere fact of exposure to fake content may have long-term consequences, influencing the perception of others (Hughes et al., 2021), or strengthening cognitive bias (Brown, 2020). Personalization is not even needed to exploit our weaknesses and “limitations for maladaptive ends” (Hughes et al., 2021).

All the examples mentioned, and this is only a selection, are gradually undermining public trust in different processes that constitute the foundations of a healthy democracy and should be understood through the prism of systemic risks and challenges. Regardless of their effectiveness, they attack basic trust at its foundation, gradually taking away any sense of stability and security. This brings us to the fundamental problem with AI-powered mis- and disinformation that relates to the psychological responses of human audiences.

PSYCHOLOGICAL AND SOCIAL EFFECTS OF AI MIS- AND DISINFORMATION

How, then, can we conceptualize the significance of AI mis- and disinformation if we are unable to assess its real quantity while at the same time being forced to recognize a significant increase and evolution of AI mis- and disinformation? Our approach proposes to shift the focus away from the *techno-fixated* conceptions about AI rightly criticized by Habgood-Coote (2023). For now, the most notable effects of AI mis- and disinformation do not seem to be rooted directly in its technological potential for amplification, increase of quality, or personalization, but in the ways users and audiences, politicians, journalists, researchers, and society at large perceive it and respond to it. For example, there are several cases in which deepfakes have already had significant real-life implications, yet these effects stem from *psychological and social reactions* of audiences to AI information manipulation rather than its pure technological features such as improved quantity or quality.

Already in 2019, (false) allegations abounded that a video message recorded by the President of Gabon was a deepfake, which triggered a military coup (Delcker, 2019). Neither quantity, quality, nor personalization played any role in this case. What was important, however, was the way that human actors *perceived* the technology and its *possible* use. This perception led to very real actions in the physical world, which must be seen as a direct outcome and effect of AI technology and its potential for misuse. Conspiracy theorists targeted real recordings of US President Joe Biden several times, suggesting, for example, that Biden was arrested or dead. In some cases, evidence from detection software of questionable quality was used as a proof of these false allegations (Horton & Sardarizadeh, 2022). These types of theories were mainly disseminated in closed environments, but they added another factor to the confusion surrounding the imagined potential of AI.

Another quite recent case demonstrates that deepfake technology can significantly distort public discourse without even being used, simply because of the reactions of human audiences. Following Hamas' attack on Israel, the Israeli government published images of child victims. These pictures were incorrectly identified as deepfakes by an AI detection software, screenshots of which were at the center of heated online debates with up to a million participants

(Maiberg, 2023a). Again, due to misled reactions of human audiences, the mere allegation that a piece of content was deepfake managed to distract and distort discourse, thus having real-world effects.

In another case, a Tesla lawyer raised doubts about the authenticity of a statement allegedly made by Elon Musk in the past, suggesting that Musk might have been a victim of a deepfake (The Guardian, 2023). This case demonstrated how the sheer existence of AI-generated media already has an influence on the perception, judgments, and possible actions of audiences by creating mistrust in the veracity of information and our ability to tell truth and facts from fakes and synthetic content. A similar line of defense was (unsuccessfully) used in the court by defendants accused of participating in the invasion on the Capitol in January 2021. Also in this case, suggestions were made that the evidence was falsified with the use of AI (Bond, 2023). For this reason, litigation experts point to a significant threat to the epistemic value of evidence and a potential challenge to tribunals around the world (Llorente, 2024).

Cases like these suggest that it is not so much (or not only) the actual use of AI for information manipulation, but the reactions of human audiences to the mere existence of AI-generated content that have powerful observable effects on the information sphere. The so-called “liar’s dividend” complemented by the “deepfake defense” (Delfino, 2023) is in fact a very strong example of how psychological reactions of human audiences to AI content produce real-life effects. And they are only indirectly connected to the actual potential and capabilities of AI, although increasing quality of synthesis may fuel a belief in the inevitability of epistemic erosion. This is partly in line with the predictions of apocalyptic visionaries, but their suggestions were mainly related to the mass occurrence of fake content, and not the mere idea of technological potential itself.

AI-powered mis- and disinformation are already affecting the perception of the information landscape, which is expressed in increased uncertainty and decreased trust in the media and news (Twomey et al., 2023). This, in turn, creates fertile ground for multiplying mis- and disinformation, especially in view of the dependencies proven in the case of disinformation in the broad sense.

EROSION OF TRUST AND SELF-FULFILLING PROPHECY

The advent of AI-generated fakes—regardless of their quantity, quality, and level of personalization—has also led to an increasing fear of AI creations. More than 70% of respondents in Germany, the UK, and the United States expressed their concerns in regard to deepfakes that circulate in the information space and over 40% of US participants had the feeling of being misled, which results in more frequent questioning of the veracity of information (Home Security Heroes, 2023; Luminate, 2023). As many as 53% of respondents in the United States believe that AI-powered mis- and disinformation will have an impact on who wins the upcoming 2024 US presidential election, and 35% were willing to admit that AI will reduce their confidence in the election results (Morning Consult, 2023). Farther afield, 71% of respondents in Poland believe that the scale of manipulation spread on the Internet via AI has increased over the last ten years, and 77% expect further increases (Mieczkowski & Kilian-Grzegorzczak, 2024).

It is all part of a broader process. AI mis- and disinformation contribute to the broader debate on fake news. The phenomenon of priming was already noticeable in this context—decreasing trust stemmed also from elite discourse and repeated warnings or speculations about the mass scale of mis- and disinformation (Van Duyn & Collier, 2019). Research by Vaccari and Chadwick (2020) showed a strong connection between the occurrence of deepfakes and increased uncertainty, which resulted in lower trust in the media. The observed dependencies on emotions prompted some authors to link the misinformation landscape with the notion of *affective polarization* (Jenke, 2023).



AI-produced synthetic content is already being confused with or ranked right next to authentic content. It seems that humans and algorithms alike are being confused, which reinforces the cumulative and erosive effect. It was visibly shown by Google's image search listing AI-generated images of historical events (Maiberg, 2023b), or Adobe Photo Stock selling AI-generated images of the military conflict in Gaza (Oremus & Verma, 2023).

None of the above-mentioned cases suggest that AI misinformation has already reached previously expected supercharging effects. Rather than fearing the one, big, powerful deep-fake that swings an election on voting day or an AI-generated image that significantly affects our perception of a war, we should pay more attention to how audiences gradually change their perceptions and reactions due to AI-fueled fears and confusion and on the effects of AI-produced content "littering" the information space. These psychological and social effects of AI mis- and disinformation are at least partly detached from its actual quantity, quality, or level of personalization, and in this context, they emphasize the importance of predictions related to the erosion of the epistemic value of content.

Sanderson and others (2024) indicate that "online misinformation [understood within our framework as AI mis- and disinformation] has been less impactful than the coverage it receives in the media," and at the same time emphasize that AI changes the "diffusion of misinformation." Like Simon and others (2023), they indicate that the original expectations may be exaggerated, but observe that AI also "changes the equation," and one of the key factors is the sheer belief of the media and public opinion that AI mis- and disinformation fundamentally change the audience's perception and have bigger potential. And this has further consequences. Esselink (2021, p. 25) links decreased trust with "psychological and epistemic consequences like denial, fear, apathy, uncertainty, or skepticism" and "hermeneutics of suspicion," whereas Hanitzsch and others (2018) with "public disenchantment" or "widespread sense of disdain."

In most of the above-mentioned cases of AI fakes it was impossible to make a definitive judgment as to actually why their authors used AI (and not other tools). In some cases, personalization, quality, and low costs might have been the reason, yet in others, AI may have been used simply because it is available or because the actors behind the fakes attach a supercharging quality to AI it may not even have. This too would suggest that psychological perceptions and reactions to AI fakes are at least as important as their technological qualities.

Poststructuralist approaches might argue that it is our technology-fixated reactions to AI (Habgood-Coote, 2023) and its alleged apocalyptic misinformation superpowers that provide the basis for these effects in the first place. In this sense, AI mis- and disinformation have real-life effects, not because of its technological features or capabilities, but because of its audience's irrational perception, hopes, and fears of the technology, which allow for them to be misled and manipulated. These fears, as Simon and others (2023) correctly assert, might contain elements of a self-fulfilling prophecy, where excessive warnings and apocalyptic narratives create the very effect they describe. However, this does not justify disregarding cases in which AI works as a tool for mis- and disinformation and affects the collective sense of security.

PSYCHOLOGICAL INOCULATION AS A SAFEGUARD?

It is an erosion of trust that seems to be the key to understanding the psychological effects of AI-powered mis- and disinformation. These do not have to be primary factors, but subsidiary and reinforce the already observed trends of increasing capabilities of mis- and disinformation, polarization, or low trust in news and the media. The latter recorded another global decline in 2023 (Newman et al., 2023). However, it should be emphasized that regaining lost trust will be extremely difficult, and as the quality of AI synthesis improves, negative processes may be reinforced and accelerated.



A comprehensive study by Twomey and others (2023) on the impact of deepfakes on epistemic trust in the context of the Russian attack against Ukraine recognized a disproportionate number of negative reactions to the technology's potential, fueled by fear and shock. It was pointed out that the dissemination of information about the potential of deepfakes translates into undermining the truth, and the deepfakes themselves “are being discussed in epistemically harmful ways” (Twomey et al., 2023, p. 16).

In some respects, these negative phenomena fit into scenarios of epistemic apocalypse, but they are not directly related to the quantity, quality, or personalization potential of AI-powered mis- and disinformation. The pessimistic visions of a post-truth society are reflected in the processes of social and political polarization and atomization. Their manifestation is the lack of shared reality, which personalization and individualization may deepen. AI-powered mis- and disinformation does not have to drive these processes, but it is undoubtedly a contributing factor. The fundamental problem is the perception of the potential of AI-powered mis- and disinformation. Simon and others (2023) point out that previous expectations in these three categories may have been exaggerated. They could be, but they did not have to be. One should note that doomsday scenarios do not concern a specific point in time and do not set clear boundaries but rather refer to the gradual erosion of trust, which leads to consequences that are difficult to quantify. Perhaps, we are just on this path. Or maybe we can get off it by using effective countermeasures.

The literature on deepfakes mentions a number of factors that enable counteracting the negative effects of misuse of technology. The repeated recommendations find partial expression in the designed regulatory frameworks for AI-generated content, which comprises—or will comprise—the combination of already mentioned watermarks, content provenance, and disclosure (AI Act, 2024). This is an important barrier that will first affect the quantity and quality of persuasiveness of AI-powered mis- and disinformation, making it more difficult to create false content and limiting its algorithmic amplification. However, this does not mean an automatic breakthrough at the epistemic and cognitive levels.

The need to significantly strengthen recipients' digital competences and social resilience has also been consistently pointed out. Far too little has been done in this regard, and the clearly visible negative consequences of poorly selected narratives not only fail to provide social immunity but even lead to further disruption of the epistemic and cognitive environment. It is significant that the recommendations for policy makers include calls to strengthen social resilience and necessity to enhance the research (Chu-Ke & Dong, 2024; Sanderson et al., 2024; Twomey et al., 2023), including studying the “social and behavioural effects of generative AI” (Sanderson et al., 2024). To understand these mechanisms, synergies among researchers, policy makers, and digital platforms are necessary—and vital.

Potential techniques comprise psychological inoculation understood as “a way to reduce susceptibility to misinformation by informing people about how they might be misinformed” (Roozenbeek et al., 2022). They might be complemented by “technocognition” or “nudging,” which may include automatic detection, tagging, and fact checking, but also modifications to the way content is presented to promote more reliable sources of information and make it easier for users to access fact checking (Gwiażdźński et al., 2022). All this collectively contributes to strengthening literacy that, among other things, requires competencies in critical evaluation and the strategic interpretation of AI and its creations and should be incorporated to education systems worldwide (Chu-Ke & Dong, 2024; Łabuz, 2024).

CONCLUSION

It seems that fears about AI-generated mis- and disinformation might be overblown, underestimated—and, above all, misunderstood. Their true potential and long-term consequences



are still not fully visible, either in terms of quantity, quality, personalization, or real-life effects. Due to the detection challenge and the rapid speed of development of AI technologies and their applications, the actual quantity of AI mis- and disinformation remains unknown. Furthermore, research so far has failed to present a quantitative or qualitative threshold AI mis- and disinformation would need to reach in order to be acknowledged as a real game changer and revolution. However, an automatic rejection of information or epistemic apocalypse narratives disregard this detection challenge completely.

Nevertheless, there is enough empirical proof for AI mis- and disinformation's existence and effects, whether the “misinformation marketplace” actually needed it or not. Fake AI images being massively spread by social media users during the conflict in Gaza demonstrate an ongoing “democratization” of disinformation technologies with every end-user being able to apply tools that used to be reserved for professional actors. Generative AI tools may be used simply because they are available at low cost or because actors project a “supercharging potential” to AI it might in fact not (yet) have. Likewise, actors of foreign influence campaigns (e.g., Russian or Chinese state actors) have also increased their use of AI for coordinated disinformation campaigns (Fredheim, 2023a, 2023b; Microsoft Threat Intelligence, 2024), which NATO StratCom does too (Bergmanis-Koräts et al., 2024).

It seems that “boosting” and “supercharging” are different from previous expectations. Claims of a pending “information apocalypse” are not so much based on present realities but on existing fears mixed with future expectations. As shown above, some of these fears have not yet materialized. However, that does not make them less real on a psychological level. On the contrary, fear might form the very basis for the most powerful real-world consequences of AI mis- and disinformation (i.e., confusion, mistrust, and uncertainty) that we can observe today. It is not that AI mis- and disinformation has proven to be so much more convincing, more cunning, or more effective, but that the advent of the new technology, its rapid development, the lack of detection methods, and our inability to distinguish between synthetic and authentic content are at both ends of a vicious circle. For the time being, psychological reactions and their social implications are responsible for the most visible and powerful consequences of deepfakes and other forms of AI mis- and disinformation.

The effects of AI-powered mis- and disinformation are still understudied. Given the creation of a regulatory framework for AI in many countries, research, common standards, frameworks, and coordination seem crucial. Shifting paradigms should include further research on the psychological and social effects of AI mis- and disinformation in various spheres of life, without an excessive focus on political processes. Contrary to warnings repeated for many years, it is not the political processes that are most at risk, but the society that co-creates political systems.

This is partly shown by statistics on the distribution of video deepfakes, where the main target is nonconsensual pornography. On the other hand, the strong emphasis on threats to political processes, repeated warnings, and fear of the technological potential of AI result in a decline in trust in the media and questioning the authenticity of information. Society feels fear and uncertainty, and policy makers are unable to propose remedies.

Therefore, AI and media literacy, awareness raising, and common standards and norms are essential. Although it has been pointed out for many years that literacy is the main barrier against AI-powered mis- and disinformation, far too little is being done in this respect, which is why society remains vulnerable and far from being inoculated. Including AI and cyberliteracy in the school curriculum is a must and politicians need to finally understand this. It might be complemented by advocating responsible narratives and discourse about the potential of AI, which in turn requires commitment and synergy with the scientific and journalistic communities.

A similarly important element will be the development of appropriate methods for detecting AI-generated content and looking for synergies between the activities of the private

and public sectors in identifying safeguards and countermeasures. This is not about technofixation, but about finding countermeasures at all stages of the AI mis- and disinformation supply chain.

Restoring trust once lost, if possible at all, seems to be a key element in overcoming the negative social and psychological implications related to the development and implementation of technologies with disruptive potential. Regardless of whether our fear is irrational or not, it is, first and foremost, real.

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CONFLICT OF INTEREST STATEMENT

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REFERENCES

- AI Act. 2024. *Proposal for Regulation of the European Parliament and of the Council Laying Down Harmonized Rules on Artificial Intelligence (Artificial Intelligence Act) and Amending Certain Union Legislative Acts*. Interinstitutional File: 2021/0106(COD). Brussels, 26 January 2024.
- Bailenson, J. N., S. Iyengar, N. Yee, and N. A. Collins. 2008. "Facial Similarity between Voters and Candidates Causes Influence." *Public Opinion Quarterly* 72(5): 935–61. <https://doi.org/10.1093/poq/nfn064>.
- Bateman, J., and D. Jackson. 2024. *Countering Disinformation Effectively: An Evidence-Based Policy Guide*. Washington, DC: Carnegie Endowment for International Peace.
- Bergmanis-Korāts, G., G. Bertolin, A. Pužule, and Y. Zeng. 2024. *AI in Support of StratCom Capabilities*. Riga: NATO StratCom COE.
- Bond, S. 2023. *People Are Trying to Claim Real Videos are Deepfakes. The Courts are not Amused*. <https://www.npr.org/2023/05/08/1174132413/people-are-trying-to-claim-real-videos-are-deepfakes-the-courts-are-not-amused>.
- Bremmer, I., and C. Kupchan. 2023. *Top Risks 2023*. New York: Eurasia Group.
- Brewster, J., M. Wang, and C. Palmer. 2023. *Plagiarism-Bot? How Low-Quality Websites Are Using AI to Deceptively Rewrite Content from Mainstream News Outlets*. <https://www.newsguardtech.com/misinformation-monitor/august-2023>.
- Bristow, T. 2023. *Keir Starmer Suffers UK Politics' First Deepfake Moment. It Won't Be the Last*. <https://www.politico.eu/article/uk-keir-starmer-labour-party-deepfake-ai-politics-elections>.
- Brown, N. 2020. "Deepfakes and the Weaponization of Disinformation." *Virginia Journal of Law & Technology* 23(1): 1–59.
- Chesney, B., and D. Citron. 2019. "Deep Fakes: A Looming Challenge for Privacy, Democracy, and National Security." *California Law Review* 107(18): 1753–1820.
- Christopher, N. 2023. *AI Modi Started as a Joke, but it Could Win Him Votes*. <https://restofworld.org/2023/ai-voice-modi-singing-politics>.
- Chu-Ke, C., and Y. Dong. 2024. "Misinformation and Literacies in the Era of Generative Artificial Intelligence: A Brief Overview and a Call for Future Research." *Emerging Media*. <https://doi.org/10.1177/27523543241240285>.
- Dekens, N. 2023. *A Practical Guide for OSINT Investigators to Combat Disinformation and Fake Reviews Driven by AI (ChatGPT)*. Cheyenne, WY: Shadow Dragon.
- Delcker, J. 2019. *Welcome to the Age of Uncertainty*. <https://www.politico.eu/article/deepfake-videos-the-future-uncertainty>.
- Delfino, R. 2023. "The Deepfake Defense—Exploring the Limits of the Law and Ethical Norms in Protecting Legal Proceedings from Lying Lawyers." *Ohio State Law Journal* 84: 1067–1124. <https://doi.org/10.2139/ssrn.4355140>.
- Diaz Ruiz, C., and T. Nilsson. 2023. "Disinformation and Echo Chambers: How Disinformation Circulates on Social Media through Identity-Driven Controversies." *Journal of Public Policy & Marketing* 42(1): 18–35. <https://doi.org/10.1177/07439156221103852>.
- Dolhansky, B., J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. C. Ferrer. 2020. *The DeepFake Detection Challenge (DFDC) Dataset*. arXiv:2006.07397. <https://doi.org/10.48550/arXiv.2006.07397>.
- Esselink, J. 2021. *Deepfakes and Extreme Beliefs: An Ethical Assessment*. Amsterdam: Vrije Universiteit Amsterdam.
- Fallis, D. 2021. "The Epistemic Threat of Deepfakes." *Philosophy & Technology* 34(4): 623–43. <https://doi.org/10.1007/s13347-020-00419-2>.

- Farid, H. 2022. "Creating, Using, Misusing, and Detecting Deep Fakes." *Journal of Online Trust and Safety* 1(4): 1–33. <https://doi.org/10.54501/jots.v1i4.56>.
- Farid, H., and H.-J. Schindler. 2020. *Deep Fakes. On the Threat of Deep Fakes to Democracy and Society*. Berlin: Konrad Adenauer Stiftung.
- Fredheim, R. 2023a. *Virtual Manipulation Brief. Generative AI and its Implications for Social Media Analysis*. Riga: NATO Strategic Communication Centre of Excellence.
- Fredheim, R. 2023b. *Virtual Manipulation Brief. Verified Propagandists and the Hamas-Israel War*. Riga: NATO Strategic Communication Centre of Excellence.
- Gold, A., and S. Fischer. 2023. *Chatbots Trigger Next Misinformation Nightmare*. <https://www.axios.com/2023/02/21/chatbots-misinformation-nightmare-chatgpt-ai>.
- Goldstein, A., J. Chao, S. Grossman, A. Stamos, and M. Tomz. 2024. "How Persuasive Is AI-Generated Propaganda?" *PNAS Nexus* 3(2): 1–7. <https://doi.org/10.1093/pnasnexus/pgae034>.
- Goldstein, J., G. Sastry, M. Musser, R. DiResta, M. Gentzel, and K. Sedova. 2023. *Generative Language Models and Automated Influence Operations: Emerging Threats and Potential Mitigations*. arXiv:2301.04246.
- Gwiaździnski, P., A. B. Gundersen, M. Piksa, I. Krysińska, J. R. Kunst, K. Noworyta, A. Olejnik, et al. 2022. "Psychological Interventions Countering Misinformation in Social Media: A Scoping Review." *Frontiers in Psychiatry* 13: 1–12. <https://doi.org/10.3389/fpsy.2022.974782>.
- Habgood-Coote, J. 2023. "Deepfakes and the Epistemic Apocalypse." *Synthese* 201: 1–23. <https://doi.org/10.1007/s11229-023-04097-3>.
- Hanitzsch, T., A. Van Dalen, and N. Steindl. 2018. "Caught in the Nexus: A Comparative and Longitudinal Analysis of Public Trust in the Press." *The International Journal of Press/Politics* 23(1): 3–23. <https://doi.org/10.1177/1940161217740695>.
- HKS Misinformation Review. 2024. *Guidelines*. <https://misinforeview.hks.harvard.edu/submit>.
- Home Security Heroes. 2023. *AI Deepfakes in 2024 Election*. <https://www.homesecurityheroes.com/ai-deepfakes-in-2024-election>.
- Home Security Heroes. 2024. *State of Deepfakes*. <https://www.homesecurityheroes.com/state-of-deepfakes/#overview-of-current-state>.
- Horton, J., and S. Sardarizadeh. 2022. *False Claims of 'Deepfake' President Biden Go Viral*. <https://www.bbc.com/news/62338593>.
- Hughes, S., O. Fried, M. Ferguson, C. Hughes, R. Hughes, X. Yao, and I. Hussey. 2021. Deepfaked Online Content Is Highly Effective in Manipulating People's Attitudes and Intentions. *PsyArXiv Preprints*: 1–6. <https://doi.org/10.31234/osf.io/4ms5a>.
- Hurst, L. 2023. *How a Fake Image of a Pentagon Explosion Shared on Twitter Caused a Real Dip on Wall Street*. <https://www.euronews.com/next/2023/05/23/fake-news-about-an-explosion-at-the-pentagon-spreads-on-verified-accounts-on-twitter>.
- Ienca, M. 2023. "On Artificial Intelligence and Manipulation." *Topoi* 42: 833–42. <https://doi.org/10.1007/s11245-023-09940-3>.
- Immerwahr, D. 2023. *What The Doomsayers Get Wrong About Deepfakes*. <https://www.newyorker.com/magazine/2023/11/20/a-history-of-fake-things-on-the-internet-walter-j-scheirer-book-review>.
- IWF. 2023. *How AI Is Being Abused to Create Child Sexual Abuse Imagery*. Cambridge: Internet Watch Foundation.
- Jenke, L. 2023. "Affective Polarization and Misinformation Belief." *Political Behavior*: 1–60. <https://doi.org/10.1007/s11109-022-09851-w>.
- Karinshak, E., and Y. Jin. 2023. "AI-Driven Disinformation: A Framework for Organizational Preparation and Response." *Journal of Communication Management* 27(4): 539–62.
- Koebler, J. 2023. *Facebook is Being Overrun with Stolen, AI-Generated Images That People Think Are Real*. <https://www.404media.co/facebook-is-being-overrun-with-stolen-ai-generated-images-that-people-think-are-real>.
- Łabuz, M. 2024. "Deepfakes Are Already Here and Already Changing the Game." In *Digital Democracy in CEE: Things to Watch for in 2024 European Parliament Elections*. Warszawa: CEE Digital Democracy Watch.
- Łabuz, M., and C. Nehring. 2024. "On the Way to Deep Fake-Democracy? Deep Fakes in Election Campaigns in 2023." *European Political Science*: 1–20. <https://doi.org/10.1057/s41304-024-00482-9>.
- Le, B., S. Tariq, A. Abuadba, K. Moore, and S. S. Woo. 2023. "Why Do Facial Deepfake Detectors Fail?" In *WDC '23: Proceedings of the 2nd Workshop on Security Implications of Deepfakes and Cheapfakes* 24–28. New York, NY: Association of Computing Machinery. <https://doi.org/10.1145/3595353.3595882>.
- Linivill, D., and P. Warren. 2023. "Infektion's Evolution: Digital Technologies and Narrative Laundering." *Media Forensics Hub Reports* 3: 1–28.
- Llorente, R. V. 2024. "Deepfakes in the Dock: Preparing International Justice for Generative AI." *The SciTech Lawyer* 20(2): 28–33.
- Lucas, J., A. Uchendu, M. Yamashita, J. Lee, S. Rohatgi, and D. Lee. 2023. "Fighting Fire with Fire: The Dual Role of LLMs in Crafting and Detecting Elusive Disinformation." In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, edited by H. Bouamor, J. Pino, and K. Bali, 14279–305. Singapore: Association for Computational Linguistics.

- Luminate. 2023. *Bots Versus Ballots: Europeans Fear AI Threat to Elections and Lack of Control over Personal Data*. <https://www.luminategroup.com/posts/news/bots-versus-ballots-europeans-fear-ai-threat-to-elections-and-lack-of-control-over-personal-data>.
- Maiberg, E. 2023a. *AI Images Detectors Are Being Used to Discredit the Real Horrors of War*. <https://www.404media.co/ai-images-detectors-are-being-used-to-discredit-the-real-horrors-of-war>.
- Maiberg, E. 2023b. *First Google Search Result for Tiananmen Square "Tank Man" Is AI Generated Selfie*. <https://404media.co/first-google-search-result-for-tiananmen-square-tank-man-is-ai-generated-selfie>.
- Martin, T. W., and D. Yoon. 2022. *These Campaigns Hope 'Deepfake' Candidates Help Get Out the Vote*. <https://www.wsj.com/articles/these-campaigns-hope-deepfake-candidates-help-get-out-the-vote-11646756345>.
- Metz, C. 2023. *'The Godfather of A.I.' Leaves Google and Warns of Danger Ahead*. <https://www.nytimes.com/2023/05/01/technology/ai-google-chatbot-engineer-quits-hinton>.
- Michaelson, R. 2023. *Turkish Presidential Candidate Quits Race after Release of Alleged Sex Tape*. <https://www.theguardian.com/world/2023/may/11/muharrem-ince-turkish-presidential-candidate-withdraws-alleged-sex-tape>.
- Microsoft Threat Intelligence. 2024. *Same Targets, New Playbooks: East Asia Threat Actors Employ Unique Methods*. Microsoft.
- Mieczkowski, P., and M. Kilian-Grzegorzczak. 2024. *Dezinformacja oczami Polaków*. Warszawa: Digital Poland.
- Mishra, A., and V. Karumbaya. 2024. *A Deadly Trifecta: Disinformation Networks, AI Memetic Warfare, and Deepfakes*. <https://gnet-research.org/2024/02/15/a-deadly-trifecta-disinformation-networks-ai-memetic-warfare-and-deepfakes>.
- Moench, M. 2024. *Taylor Swift Searches Blocked by X Amid Circulation of Deepfakes*. <https://time.com/6589487/taylor-swift-searches-blocked-x-twitter-deepfakes-response>.
- Morning Consult. 2023. *National Tracking Poll*. https://pro-assets.morningconsult.com/wp-uploads/2023/09/2308055_topline_AXIOS_AI_Adults_v1_EP-1.pdf.
- Motta, M., J. Hwang, and D. Stecula. 2023. "What Goes Down Must Come Up? Pandemic-Related Misinformation Search Behavior during an Unplanned Facebook Outage." *Health Communication*: 1–12. <https://doi.org/10.1080/10410236.2023.2254583>.
- Muñoz, K. 2024. "The AI Election Year: How to Counter the Impact of Artificial Intelligence." *DGAP Memo* 1: 1–4.
- Murphy, G., D. Ching, J. Twomey, and C. Linehan. 2023. "Face/Off: Changing the Face of Movies with Deepfakes." *PLoS One* 18: e0287503. <https://doi.org/10.1371/journal.pone.0287503>.
- Murphy, H. 2024. *Audio Deepfakes Emerge as Weapon of Choice in Election Disinformation*. <https://ft.com/content/bd75b678-044f-409e-b987-8704d6a704ea>.
- Newman, N., R. Fletcher, K. Eddy, C. T. Robertson, and R. K. Nielsen. 2023. *Reuters Institute Digital News Report 2023*. Oxford: Reuters Institute for the Study of Journalism.
- Newport, C. 2023. *What Kind of Mind Does ChatGPT have?* <https://www.newyorker.com/science/annals-of-artificial-intelligence/what-kind-of-mind-does-chatgpt-have>.
- Nicas, J. 2023. *Is Argentina the First A.I. Election?* <https://www.nytimes.com/2023/11/15/world/americas/argentina-election-ai-milei-massa.html>.
- Nilesh, C. 2020. "We've Just Seen the First Use of Deepfakes in an Indian Election Campaign." *Vice*. <https://www.vice.com/en/article/jgedjb/the-first-use-of-deepfakes-in-indian-election-by-bjp>.
- Okolie, C. 2023. "Artificial Intelligence-Altered Videos (Deepfakes), Image-Based Sexual Abuse, and Data Privacy Concerns." *Journal of International Women's Studies* 25(2): 1–16.
- Olszanowski, M., P. Lewandowska, A. Ozimek, and N. Frankowska. 2022. "The Effect of Facial Self Resemblance on Emotional Mimicry." *Journal of Nonverbal Behavior* 46(2): 197–213. <https://doi.org/10.1007/s10919-021-00395-x>.
- Onfido. 2024. *Identity Fraud Report 2024*. London: Onfido.
- Oremus, W., and P. Verma. 2023. *These Look Like Prizewinning Photos. They're AI Fakes*. <https://www.washingtonpost.com/technology/2023/11/23/stock-photos-ai-images-controversy>.
- Osmundsen, M., A. Bor, P. B. Vahlstrup, A. Bechmann, and M. B. Petersen. 2021. "Partisan Polarization Is the Primary Psychological Motivation behind Political Fake News Sharing on Twitter." *American Political Science Review* 115(3): 999–1015. <https://doi.org/10.1017/S0003055421000290>.
- Oyserman, D., and A. Dawson. 2020. "Your Fake News, Our Facts: Identity-Based Motivation Shapes What We Believe, Share, and Accept." In *The Psychology of Fake News*, edited by R. Greifeneder, M. Jaffe, E. Newman, and N. Schwarz, 173–95. London: Routledge.
- Rini, R. 2020. "Deepfakes and the Epistemic Backstop." *Philosophers' Imprint* 20(24): 1–16.
- Rödter, T. 2023. *Das Geschäft mit der geklonten Stimme*. <https://www.tagesschau.de/investigativ/swr/ki-kuenstlich-e-intelligenz-voice-cloning-100.html>.
- Roozenbeek, J., S. van der Linden, B. Goldberg, S. Rathje, and S. Lewandowsky. 2022. "Psychological Inoculation Improves Resilience against Misinformation on Social Media." *Science Advances* 8(34): 1–11. <https://doi.org/10.1126/sciadv.abo6254>.
- Sadasivan, V., A. Kumar, S. Balasubramanian, W. Wang, and S. Feizi. 2023. *Can AI-Generated Text be Reliably Detected?* arXiv:2303.11156.

- Sanderson, Z., S. Messing, and J. A. Tucker. 2024. *Misunderstood Mechanics: How AI, TikTok, and the Liar's Dividend Might Affect the 2024 Elections*. <https://www.brookings.edu/articles/misunderstood-mechanics-how-ai-tiktok-and-the-liars-dividend-might-affect-the-2024-elections>.
- Sanger, D. E., and S. L. Myers. 2023. *China Sows Disinformation about Hawaii Fires Using New Techniques*. <https://www.nytimes.com/2023/09/11/us/politics/china-disinformation-ai.html>.
- Schick, N. 2020. *Deep Fakes and the Infocalypse*. Ottawa: Octopus Books.
- Simchon, A., M. Edwards, and S. Lewandowsky. 2024. "The Persuasive Effects of Political Microtargeting in the Age of Generative AI." *PNAS Nexus* 3: 1–5. <https://doi.org/10.1093/pnasnexus/pgae035>.
- Simon, F. M., S. Altay, and H. Mercier. 2023. "Misinformation Reloaded? Fears about the Impact of Generative AI on Misinformation are Overblown." *Harvard Kennedy School Misinformation Review* 4(5). <https://doi.org/10.37016/mr-2020-127>.
- Simon, S. 2023. *Opinion: Did He Really Say That?* <https://www.npr.org/2023/10/21/1207749595/opinion-did-he-really-say-that>.
- Spitale, G., N. Biller-Andorno, and F. Germani. 2023. "AI Model GPT-3 (Dis)Informs Us Better than Humans." *Science Advances* 9(26): 1–9. <https://doi.org/10.1126/sciadv.adh1850>.
- Tang, R., Y.-N. Chuang, and X. Hu. 2023. *The Science of Detecting LLM-Generated Texts*. arXiv:2303.07205.
- The Guardian. 2023. *Elon Musk's Statements Could Be 'Deepfakes,' Tesla Defence Lawyers Tell Court*. <https://www.theguardian.com/technology/2023/apr/27/elon-musks-statements-could-be-deepfakes-tesla-defence-lawyers-tell-court>.
- Tong, A., and H. Coster. 2023. *Meet Ashley, the World's First AI-Powered Political Campaign Caller*. <https://www.reuters.com/technology/meet-ashley-worlds-first-ai-powered-political-campaign-caller-2023-12-12>.
- Twomey, J., D. Ching, M. P. Aylett, M. Quayle, C. Linehan, and G. Murphy. 2023. "Do Deepfake Videos Undermine Our Epistemic Trust? A Thematic Analysis of Tweets that Discuss Deepfakes in the Russian Invasion of Ukraine." *PLoS One* 18(10): 1–22. <https://doi.org/10.1371/journal.pone.0291668>.
- Vaccari, C., and A. Chadwick. 2020. "Deepfakes and Disinformation: Exploring the Impact of Synthetic Political Video on Deception, Uncertainty, and Trust in News." *Social Media + Society* 6(1): 205630512090340. <https://doi.org/10.1177/2056305120903408>.
- Van Duyn, E., and J. Collier. 2019. "Priming and Fake News: The Effects of Elite Discourse on Evaluations of News Media." *Mass Communication and Society* 22(1): 29–48. <https://doi.org/10.1080/15205436.2018.1511807>.
- World Economic Forum. 2024. *The Global Risks Report 2024*. Geneva: World Economic Forum.
- Yang, K.-C., and F. Menczer. 2023. *Anatomy of an AI-Powered Malicious Social Botnet*. arXiv:2307.16336.
- Yang, K.-C., D. Singh, and F. Menczer. 2024. *Characteristics and Prevalence of Fake Social Media Profiles with AI-Generated Faces*. arXiv:2401.02627v1.
- Yu, J. 2024. *Deepfake Video Call Scams Global Firm out of \$26 Million: SCMP*. <https://www.bloomberg.com/news/articles/2024-02-04/deepfake-video-call-scams-global-firm-out-of-26-million-scmp>.
- Zarouali, B., T. Dobber, G. De Pauw, and C. de Vreese. 2022. "Using a Personality-Profiling Algorithm to Investigate Political Microtargeting: Assessing the Persuasion Effects of Personality-Tailored Ads on Social Media." *Communication Research* 49(8): 1066–91. <https://doi.org/10.1177/0093650220961965>.

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